La Scuola d'Italia / Dr. Huson / IB Math 12
2 November 2026

Nickname:

3.1 Classwork: The Unit Circle and Trigonometric Graphs (no calculator)

All angles are in radians. Give exact answers.

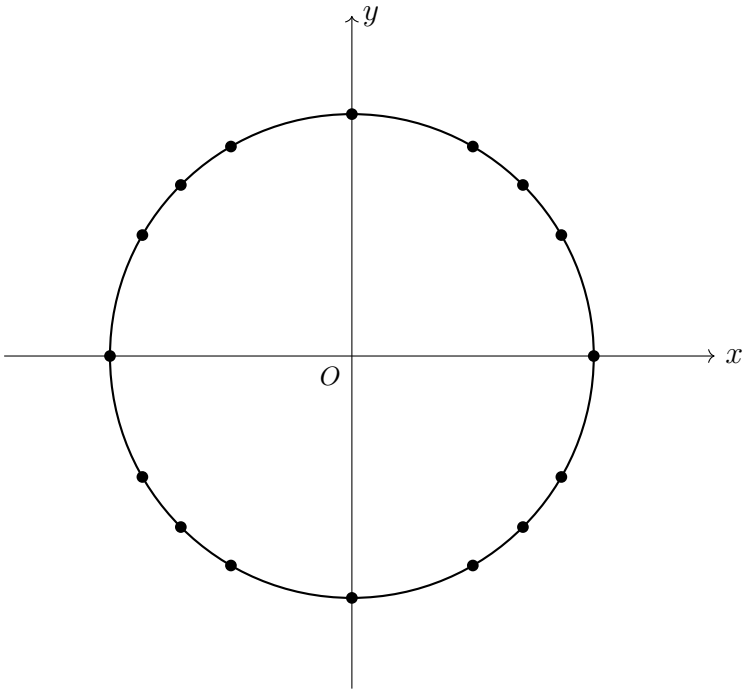
1. The diagram in the box shows the unit circle with 16 points marked.

   (a) From memory, write the angle θ in radians next to each point, inside the circle, and its coordinates $(\cos\theta,\ \sin\theta)$ outside the circle.

   (b) Write down the exact value of $\sin\dfrac{5\pi}{6}$, $\cos\dfrac{4\pi}{3}$ and $\tan\dfrac{7\pi}{4}$.

   (c) Verify the identity $\sin^2\theta + \cos^2\theta = 1$ for $\theta = \dfrac{5\pi}{6}$.

   (d) Given that $\sin\theta = \dfrac{2}{3}$ and $\dfrac{\pi}{2} < \theta < \pi$, find the exact value of $\cos\theta$.



2. Use your values from Problem 1. You may use $\dfrac{\sqrt{3}}{2} \approx 0.87$ and $\dfrac{\sqrt{2}}{2} \approx 0.71$.

   (a) On the grid, plot $y = \sin x$ at $x = 0,\ \dfrac{\pi}{6},\ \dfrac{\pi}{3},\ \ldots,\ 2\pi$. Join the points with a smooth curve and label it $y = \sin x$.

   (b) On the same grid, plot and label $y = \cos x$ in the same way.

   The point $P(\cos\theta,\ \sin\theta)$ lies on the unit circle, and the line OP makes an angle θ with the positive x-axis.

   (c) Show that the gradient of the line OP is $\tan\theta$.

   (d) The line L_1 makes an angle of $\dfrac{\pi}{6}$ with the positive x-axis and has y-intercept 2. Find the equation of L_1.

   (e) The line L_2 makes an angle of $\dfrac{2\pi}{3}$ with the positive x-axis and passes through the point $(1,\ 4)$. Find the equation of L_2.

